DATASET OVERVIEW

Rows: 478

Columns: 22

Data types:

brand object
model object
top_speed_kmh int64
battery_capacity_kWh float64
battery_type object
number_of_cells float64
torque_nm float64
efficiency_wh_per_km int64
range_km int64
acceleration_0_100_s float64
fast_charging_power_kw_dc float64
fast_charge_port object
towing_capacity_kg float64
cargo_volume_l object
seats int64
drivetrain object
segment object
length_mm int64
width_mm int64
height_mm int64
car_body_type object
source_url object
dtype: object

DATA CLEANING

Duplicate rows: 0

Missing values before preprocessing:

```
model                    1
number_of_cells          202
torque_nm                7
fast_charging_power_kw_dc    1
fast_charge_port         1
towing_capacity_kg       26
cargo_volume_l           1
dtype: int64
```

Columns removed:

['cargo_volume_l', 'source_url', 'battery_type', 'number_of_cells', 'model', 'fast_charge_port']

Dataset shape after cleaning/feature selection: (478, 17)

EXPLORATORY DATA ANALYSIS

Four EDA visualizations saved in: eda_plots

HYPOTHESIS TESTING

TEST 1 - Pearson Correlation

H0: There is no linear correlation between battery capacity and range (rho = 0).

H1: There is a linear correlation between battery capacity and range (rho != 0).

Correlation (r): 0.8804

p-value: 2.30714e-156

Conclusion: Reject H0 at the 5% significance level.

TEST 2 - Welch Independent Samples t-test

H0: The mean range of SUVs and Sedans is equal.

H1: The mean range of SUVs and Sedans is different.

SUV mean: 396.52 km

Sedan mean: 496.83 km

t-statistic: -8.9138

p-value: 6.77019e-14

Conclusion: Reject H0 at the 5% significance level.

MACHINE LEARNING

Training records: 382

Testing records: 96

Model Evaluation Results:

Model	R2	MAE_km	RMSE_km
Linear Regression	0.96953	14.429089	17.956093
Random Forest	0.96321	14.308438	19.730776

Cleaned dataset saved as: electric_vehicles_cleaned.csv

Model results saved as: model_results.csv